Energy-Efficient, Risk-Aware Control of Domestic Hot Water Systems Under Stochastic Demand

Anonymous Author(s)

Abstract

Domestic hot-water (DHW) storage tanks are flexible electrical loads, yet are commonly controlled using fixed thermostats or static timers that ignore uncertainty in user demand and incur unnecessary energy losses. We study adaptive control of an electric immersion heater under stochastic hot-water usage, with an explicit focus on the value of probabilistic demand information. User demand is modelled as a non-homogeneous Poisson process whose time-varying intensity $\lambda(t)$ captures diurnal usage patterns and serves as a deployable demand-risk signal.

We formulate heater control in a physics-based simulation that captures realistic thermal dynamics, duration-aware hot-water draws, and binary actuation with dwell-time constraints. A Proximal Policy Optimisation (PPO) agent is trained to balance energy consumption against volume-weighted comfort violations. We benchmark PPO against deployable baseline controllers, including thermostatic, timer-based, and a risk-aware heuristic that exploits the same $\lambda(t)$ signal. Results show that access to demand-risk information delivers most of the achievable energy–comfort gains, while reinforcement learning provides only incremental refinement.

CCS Concepts

• **Computing methodologies** → **Reinforcement learning**; • **Computer systems organization** → **Embedded and cyber-physical systems**; • **Applied computing** → *Multi-criterion optimization and decision-making*; • **Hardware** → *Energy generation and storage.*

Keywords

energy efficiency, stochastic user demand, reinforcement learning, domestic hot-water systems

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1 Introduction

Residential buildings account for a substantial share of global energy consumption, with domestic hot-water (DHW) provision representing a significant and largely inflexible load in many households. As heating systems electrify and variable renewable generation increases, there is growing interest in shifting flexible electrical demand in time. Electric immersion heaters coupled with hot-water storage tanks are well suited to this role: they store thermal energy, exhibit strong daily usage regularities, and can often defer heating without immediately affecting user experience.

Despite this flexibility, most domestic immersion heaters are controlled by simple thermostatic hysteresis or fixed timer schedules. Thermostats maintain the tank temperature within a narrow band regardless of anticipated demand, leading to unnecessary reheating and standing losses during idle periods. Timers reduce energy use by restricting heating to predefined windows but are brittle under day-to-day variability and irregular routines. Both strategies treat hot-water demand as either fully predictable or entirely exogenous, and neither explicitly reasons about uncertainty.

A central challenge in DHW control is that user demand is structured but stochastic. Hot-water usage exhibits repeatable diurnal patterns, such as morning and evening peaks, yet the exact timing, duration, and volume of events vary substantially from day to day. Controllers that ignore this uncertainty tend to either maintain overly conservative temperature buffers, wasting energy, or to under-prepare and incur comfort failures during prolonged or atypical demand. Effective control therefore requires anticipating demand risk while relying only on information that is realistically available at runtime.

Recent work has explored reinforcement learning (RL) for DHW and building energy control, motivated by the promise of learning adaptive policies without explicit forecasting or system identification. However, a recurring limitation in this literature is that learned controllers are often compared only against weak baselines, such as static timers or fixed thermostats. When RL agents are provided with additional information, such as occupancy cues or probabilistic forecasts, it becomes unclear whether observed gains arise from the learning algorithm itself or simply from access to better information.

In this paper, we adopt a value-of-information perspective. We ask whether access to a probabilistic demand-risk signal is sufficient to achieve most of the attainable energy savings, or whether a complex learning-based controller provides substantial additional benefit beyond a simple, interpretable heuristic. To address this question, we model hot-water demand as a non-homogeneous Poisson process (NHPP) with a time-varying intensity $\lambda(t)$ that captures diurnal structure. We interpret $\lambda(t)$ as a deployable risk signal available at runtime and provide it to both a reinforcement learning agent and a rule-based controller.

We formulate immersion-heater control as a continuing decision-making problem in a physics-based simulation environment. The environment captures heat input, standing losses, and the thermal impact of hot-water draws using a duration-aware demand model in which each usage event persists over multiple time steps. The controller observes only the tank temperature, time-of-day

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features, the instantaneous demand intensity $\lambda(t)$, and the current aggregate draw rate. Actuation is restricted to binary on/off control with minimum dwell-time constraints to reflect realistic hardware operation.

We evaluate a Proximal Policy Optimisation (PPO) agent against four deployable baselines: always-on operation, fixed-setpoint thermostat control, a night-off timer schedule, and a risk-aware heuristic that adjusts its temperature target based on $\lambda(t)$. All controllers are evaluated under identical stochastic demand realisations and compared using engineering-relevant metrics, including weekly energy consumption, volume-weighted unmet demand, and switching frequency.

This paper makes three contributions. First, we introduce a duration-aware DHW control environment that combines stochastic demand, physically consistent thermal dynamics, and realistic actuation constraints while exposing only deployable observations. Second, we present a rigorous evaluation protocol that benchmarks reinforcement learning against strong, interpretable baselines, including a risk-aware heuristic that uses the same probabilistic information as the learned policy. Third, we provide an empirical analysis of energy–comfort trade-offs showing that much of the performance gain arises from access to demand-risk information itself, with reinforcement learning offering incremental refinement rather than a qualitative shift in achievable performance.

The remainder of the paper describes the environment and controller design (Section 3), presents comparative results and trade-off analyses (Section 4), and discusses implications for deployable domestic hot-water control and demand-side flexibility (Section 5).

2 Related Work

Our work lies at the intersection of (i) stochastic domestic hot-water (DHW) demand modelling and (ii) deployable control under uncertainty for electric storage water heaters. Prior work spans simple rule-based controllers, Model Predictive Control (MPC), and Reinforcement Learning (RL). However, a recurring evaluation pitfall is that RL is often compared only against weak baselines (e.g., fixed timers), making it unclear whether reported gains come from the learning algorithm or simply from access to better information. We therefore adopt a value-of-information framing: given a probabilistic risk signal for demand (here, an NHPP intensity), is a complex RL policy necessary, or can an interpretable risk-aware heuristic capture most of the achievable benefit?

2.1 Stochastic Modelling of DHW Demand and Uncertainty

A consistent finding in DHW studies is that usage is structured yet uncertain: diurnal regularities (e.g., morning/evening peaks) coexist with substantial day-to-day variability. This motivates stochastic demand models rather than deterministic schedules. De Coninck et al. generate hot-water events through occupancy-driven stochastic modelling (Markov-style behaviour with survival analysis), producing clustered bursts that resemble realistic household routines [2].

More recent work emphasises data-driven probabilistic simulators that capture user heterogeneity and temporal effects (season/weekday), showing that personalised stochastic models can substantially reduce modelling error relative to older static assumptions [5].

A common abstraction in this line of work is the Non-Homogeneous Poisson Process (NHPP), where a time-varying intensity $\lambda(t)$ represents time-of-day-dependent demand likelihood. Our use of NHPP aligns with this literature, but we explicitly interpret $\lambda(t)$ as a deployable risk signal that can be consumed by either heuristics or learning-based controllers.

2.2 Rule-Based and Risk-Aware Heuristics

Despite interest in "smart" control, many practical deployments favour rule-based approaches due to transparency and low integration cost. A broad class of heuristics modulates heater behaviour using time-of-day, occupancy cues, tariffs, or renewable availability. Such strategies effectively perform risk buffering: they maintain more thermal headroom before likely usage and allow temperatures to drift lower during low-risk periods to reduce standing losses. Examples include rule-based setpoint reductions during unoccupied or high-cost periods and resumption of heating ahead of expected demand windows, as demonstrated in demand-side management contexts [2]. Related scheduling-style approaches incorporate consumption habits and (sometimes) dynamic prices while attempting to preserve comfort [7].

This motivates our Risk-Band baseline: an interpretable controller that uses the same risk signal $\lambda(t)$ available to RL, switching between lower and higher temperature targets. The purpose is not to "beat timers", but to test whether risk information alone is sufficient for strong performance.

2.3 Model Predictive Control and Chance-Constrained Variants

MPC is a natural benchmark for DHW because it explicitly exploits thermal inertia and can incorporate constraints such as minimum ON/OFF times and cycling limits. When accurate models and forecasts are available, MPC often performs near-optimally and provides predictable comfort guarantees. However, the literature repeatedly notes that practical MPC deployment can be expensive due to system identification, state estimation, and forecasting requirements, and performance can degrade under demand uncertainty or forecast mismatch. Chance-constrained and probabilistic MPC variants address uncertainty by maintaining buffers to satisfy demand with high probability (e.g., via quantile-based demand profiles), conceptually close to the risk-buffering behaviour we aim to study [3].

Our work targets a lighter-weight "model-free / low-model" niche: instead of requiring an optimisation loop and an explicit forecast horizon, we compare a simple risk-aware heuristic and an RL agent operating on the same deployable observations.

2.4 Reinforcement Learning for Water-Heating Control

RL has been widely explored for DHW control as a way to learn policies from interaction, potentially avoiding explicit modelling and forecasting. Ruelens et al. applied batch RL for water heater control and reported cost reductions relative to thermostat baselines [6].

Deep RL approaches have also been evaluated in broader building and water heating contexts, reporting energy savings while attempting to maintain comfort [1].

However, multiple studies and surveys emphasise challenges that are particularly relevant to your setting: (i) partial observability due to uncertain occupant demand, (ii) sparse, high-impact comfort violations ("cold shower" events) that destabilise learning, and (iii) binary actuation with dwell-time constraints that creates high-variance credit assignment and can promote degenerate policies (over-conservative "always hot" or under-heating with occasional failures). These limitations are discussed in simulation studies and reviews of RL for building energy control [4, 8].

We therefore design a duration-aware demand model (multi-step draws) and enforce actuation realism (dwell times and switching penalties) so that learned policies are evaluated under constraints that mirror practical heaters.

2.5 Benchmarking Gaps and the Value-of-Information Framing

A key gap highlighted across comparative studies is that RL is often benchmarked against weak baselines (e.g., naive timers), which can overstate the value of "RL" when the true advantage comes from additional information (forecasts, occupancy cues, engineered features). Comprehensive benchmarks that include strong heuristics and MPC-style baselines are increasingly used to clarify when RL provides incremental benefit [4, 8].

Our contribution follows this best-practice direction: we introduce a Risk-Band Controller that uses the same NHPP risk signal $\lambda(t)$ as the PPO agent. This isolates (a) the benefit of providing probabilistic demand information from (b) any additional benefit of policy optimisation via RL. In other words, we ask an engineering question: does RL materially improve the energy–comfort trade-off beyond what a transparent risk-threshold heuristic can already achieve? This positioning aligns with the broader "right tool for the job" theme in DHW control research, where simplicity, safety, and deployability are first-class concerns.

3 Methodology

We study control of an electric immersion heater feeding a domestic hot-water (DHW) storage tank under stochastic user demand. The aim is to reduce electrical energy consumption while maintaining adequate delivered hot-water temperature during usage, using only information that is realistically available at runtime. A key design choice in this paper is a value-of-information framing: given access to a probabilistic demand-risk signal, we test whether a simple rule-based controller can exploit this information nearly as well as a learned reinforcement learning (RL) policy.

We implement a weekly, continuing-control environment with (i) stochastic demand start times and multi-step draw durations, (ii) physically consistent tank thermal dynamics, and (iii) an energy–comfort objective used for RL training. Learned and non-learned controllers are evaluated under identical demand realisations using fixed random seeds.

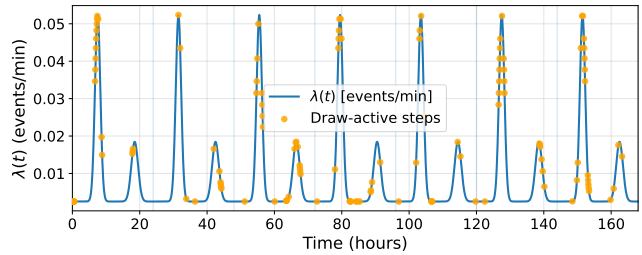

Figure 1: Example weekly demand-risk profile from the NHPP model. The curve shows the time-varying intensity $\lambda(t)$ and markers indicate steps where draw is active in that realisation. This visualises the deployable risk signal used by Risk-Band and provided as an observation to PPO.

3.1 Stochastic demand model with duration

Hot-water usage is modelled in discrete time with step size $\Delta t = 5$ min. Demand event starts are generated by a Non-Homogeneous Poisson Process (NHPP) with time-varying intensity $\lambda(t)$ (events min^{-1}) that captures diurnal structure:

$$\lambda(t) = \lambda_{\text{base}} + \sum_{j \in \{m,e\}} A_j \exp\left(-\frac{(t-\mu_j)^2}{2\sigma_j^2}\right), \tag{1}$$

where λ_{base} is a background rate and the two Gaussian components represent morning and evening peaks.

At each step t, the number of new event starts k_t is sampled from a Poisson distribution with mean $\mu_t = \lambda(t)\Delta t$. Each started event i is assigned a type $z \in \{\text{handwash}, \text{shower}\}$ and is parameterised by a total volume V_i and duration D_i sampled from type-specific distributions. To represent sustained consumption (e.g., a shower spanning multiple steps) rather than instantaneous impulses, each event induces a constant per-step draw over its active duration. Let $n_i = \lceil D_i/\Delta t \rceil$ be the number of steps, and define the per-step volume for that event as $q_i = V_i/n_i$. The environment maintains a queue of active events $\mathcal{A}_t$ and the aggregate draw per step is

$$Q_t = \sum_{i \in \mathcal{A}_t} q_i \quad (\text{L/step}). \tag{2}$$

We interpret $\lambda(t)$ as a deployable demand-risk signal available at runtime (e.g., derived from a learned diurnal profile). This signal is provided to both the learned policy and the risk-aware heuristic baseline, enabling direct comparison of algorithmic benefit beyond information availability.

3.2 Environment formulation

We model the system as a continuing Markov decision process and truncate rollouts to one simulated week (2016 steps) for logging, training updates, and evaluation. The state evolves under heater actuation and demand-driven mixing dynamics.

3.2.1 Thermal dynamics. The DHW tank is represented as a single, fully mixed thermal node. Let T_t denote bulk water temperature (in °C) at the start of step t. The controller selects a binary heater command $a_t \in \{0, 1\}$ (OFF/ON). Heater power input and standing

Table 1: Physical and control parameters used in all simulations.

Category	Parameter	Value
Tank	Water mass m	150 kg
Tank	Specific heat c_p	4186 $\mathrm{J\,kg^{-1}\,K^{-1}}$
Heater	Rated power P_{heater}	3 kW
Heater	Efficiency η	1.0
Losses	Loss coefficient U_A	15 $\mathrm{W\,K^{-1}}$
Boundary	Mains temperature T_{main}	10 °C
Boundary	Ambient temperature T_{amb}	15 °C
Comfort	Target temperature T_{target}	50 °C
Safety	Maximum temperature $T_{\max}$	65 °C
Time	Timestep Δt	5 min
Control	Min ON time d_{on}	15 min
Control	Min OFF time d_{off}	15 min

losses are integrated via forward Euler:

$$T_t^+ = T_t + \frac{\eta P_{\text{heater}} \tilde{a}_t - U_A(T_t - T_{\text{amb}})}{m c_p} \Delta t_s, \tag{3}$$

where m is water mass, c_p is specific heat capacity, P_{heater} is heater power, η is efficiency, U_A is the standing-loss coefficient, T_{amb} is ambient temperature, and Δt_s is step duration in seconds. The symbol $\tilde{a}_t$ denotes the applied action after enforcing dwell-time constraints (Section 3.2.2).

Hot-water demand then causes mixing with mains water at temperature T_{main}. Under the fully mixed approximation, drawing Q_t litres during a step replaces an equivalent volume with mains inflow. Let $f_t = \min(Q_t/V_{\text{tank}}, 1)$ be the replaced fraction of the tank volume V_{tank} (litres). The post-draw temperature is

$$T_{t+1} = T_t^+ - f_t(T_t^+ - T_{\text{main}}). \tag{4}$$

We define the delivered temperature during step t as

$$T_t^{\text{del}} = T_t^+, \tag{5}$$

which represents the water temperature available at the point of delivery before dilution due to replacement. Delivered temperature is defined as the tank temperature immediately prior to draw-induced mixing in each step. Comfort penalties are proportional to both unmet draw volume and temperature shortfall. This formulation captures both severity and scale of discomfort while remaining compatible with a single-node, fully mixed tank model. We do not explicitly model outlet-layer stratification, and therefore interpret absolute failure rates cautiously while focusing on comparative controller performance.

3.2.2 Action constraints. To reflect relay limitations and prevent chattering, we impose minimum dwell times. After an ON (or OFF) transition, the applied heater state is locked for d_{on} (or d_{off}) minutes, corresponding to $k_{\text{on}} = \lceil d_{\text{on}}/\Delta t \rceil$ and $k_{\text{off}} = \lceil d_{\text{off}}/\Delta t \rceil$ steps. The environment masks the requested action a_t to produce $\tilde{a}_t$ that satisfies these constraints.

3.2.3 Observation and partial observability. The observation is designed to be deployable using standard sensors and time features:

$$o_t = \left[\, T_t,\ \sin(\omega t),\ \cos(\omega t),\ \lambda(t),\ Q_t \,\right], \tag{6}$$

where $\omega = 2\pi/24\,\mathrm{h}^{-1}$. The queue of active events and their remaining durations are not observed, so the control problem is partially observable. Controllers must manage thermal reserves without explicit knowledge of future demand or how long ongoing draws will persist.

3.3 Reward design for RL training

The RL agent is trained to minimise energy usage while avoiding low delivered temperatures during demand. Let T_{target} be the comfort threshold. The instantaneous temperature deficit is

$$\Delta T_t = \max(0, T_{\text{target}} - T_t^{\text{del}}). \tag{7}$$

The per-step reward is a weighted sum of energy cost, comfort penalty during draws, a risk-shaped term, and switching penalty:

$$r_t = r_t^{\text{energy}} + r_t^{\text{comfort}} + r_t^{\text{shape}} + r_t^{\text{switch}}. \tag{8}$$

Energy use is penalised when the heater is active:

$$r_t^{\text{energy}} = -w_{\text{energy}} c_e E_t, \tag{9}$$

where c_e is electricity price and E_t is step energy in kWh.

Comfort is penalised only during demand, using a volume-weighted deficit:

$$r_t^{\text{comfort}} = -w_{\text{fail}} Q_t \Delta T_t, \tag{10}$$

and we set any positive reward for meeting demand to zero in our primary experiments to keep the objective as cost minimisation.

To encourage anticipatory heating in high-risk periods without revealing future demand, we include a shaping term that scales the deficit by the expected number of demand starts $\mu_t = \lambda(t)\Delta t$:

$$r_t^{\text{shape}} = -w_{\text{expect}} \mu_t \Delta T_t. \tag{11}$$

Switching is discouraged to reflect actuator wear:

$$r_t^{\text{switch}} = -w_{\text{switch}} \mathbb{1}[\tilde{a}_t \neq \tilde{a}_{t-1}]. \tag{12}$$

The shaping term uses the same risk signal $\lambda(t)$ that is also made available to controllers at runtime. In evaluation, baselines do not use reward shaping instead they are assessed only by physical outcomes (energy use, unmet volume, and switching).

3.4 Controllers

We evaluate five controllers. All controllers act through the same binary heater command and are subject to the same dwell-time constraints in the environment.

3.4.1 Always-On. Always-On continuously requests heating at every step, i.e., $a_t = 1$, subject to the environment's physical limits and safety clamp at $T_{\max}$. It is included as an extreme reference strategy rather than a deployable baseline. Comfort shortfalls may still occur under sustained draws due to finite heater power. Energy consumption under Always-On should therefore be interpreted as a high-consumption reference under aggressive control, not as realistic household operation.

3.4.2 Thermostat. The Thermostat controller is a fixed-setpoint hysteresis rule with setpoint T_{set} and deadband δ. Let $u_t \in \{0, 1\}$ be the internal heater state. The rule is:

$$u_t = \begin{cases} 1 & \text{if } u_{t-1} = 0 \text{ and } T_t \leq T_{\text{set}} - \delta \\ 0 & \text{if } u_{t-1} = 1 \text{ and } T_t \geq T_{\text{set}} + \delta \\ u_{t-1} & \text{otherwise.} \end{cases} \tag{13}$$

The requested action is $a_t = u_t$.

3.4.3 Timer. Timer applies a fixed night-off schedule and otherwise reverts to the Thermostat rule. Let t be the step index and define the step-of-day $s_t = t \bmod N_{\text{day}}$ with $N_{\text{day}} = 24\text{h}/\Delta t$. The heater is forced OFF in a predefined window (e.g., 23:00–06:00):

$$a_t = \begin{cases} 0 & \text{if } s_t \in \mathcal{W}_{\text{off}} \\ u_t & \text{otherwise,} \end{cases} \tag{14}$$

where u_t is the Thermostat decision.

3.4.4 Risk-Band. Risk-Band is an interpretable risk-aware heuristic that uses the instantaneous NHPP intensity $\lambda(t)$ to switch between a low and high setpoint:

$$T_{\text{set}}(t) = \begin{cases} T_{\text{high}} & \text{if } \lambda(t) \geq \lambda_{\text{thresh}} \\ T_{\text{low}} & \text{otherwise.} \end{cases} \tag{15}$$

A hysteresis rule identical to Section 3.4.2 is then applied using $T_{\text{set}}(t)$ in place of a fixed setpoint. This baseline shares the same risk signal as the RL agent and directly tests whether the signal can be exploited effectively with simple logic.

3.4.5 PPO policy. We train a stochastic policy using Proximal Policy Optimisation (PPO) on observations o_t (Section 3.2.3). The policy outputs a binary action $a_t \in \{0, 1\}$ and is trained under the reward in Section 3.3. The policy uses a standard multilayer perceptron with two hidden layers and is trained with two-week rollouts using four parallel environments. We use a discount factor $\gamma = 0.999$, learning rate 3×10^{-4}, batch size 256, and 10 epochs per update, with observation normalisation (VecNormalize) enabled and reward normalisation disabled. Evaluation uses the deterministic policy mode.

Hyperparameters largely follow standard PPO defaults and were not extensively tuned. This is a deliberate design choice: the objective of the study is not to maximise PPO performance, but to assess the marginal benefit of reinforcement learning relative to information-matched heuristic baselines when both have access to the same demand-risk signal $\lambda(t)$. Minor adjustments (e.g., rollout length) were made only to align with the weekly horizon and dwell-time constraints of the environment.

3.5 Evaluation protocol and metrics

We evaluate each controller for one simulated week under N independent demand realisations generated by fixed random seeds. For each seed, all controllers face identical event start counts, event volumes, and event durations. Reported performance is aggregated as mean and standard deviation over seeds.

Baseline controllers use fixed parameters selected once from typical domestic operating ranges and are held constant across all evaluation seeds. No baseline parameters are adapted online or tuned per-seed. This ensures that differences reflect controller structure and information access (e.g., $\lambda(t)$), rather than per-instance retuning.

We report engineering-relevant weekly metrics:

- Total energy consumption (kWh/week), $\sum_t E_t$.
- Duty cycle, the fraction of steps where $\tilde{a}_t = 1$.
- Switch count (per week), the number of changes in applied action $\tilde{a}_t$.
- Total draw volume (L/week), $\sum_t Q_t$.
- Unmet volume (L/week), $\sum_t U_t$, where $U_t = Q_t \mathbb{1}[\Delta T_t > 0]$.
- Unmet volume fraction, $F_{\text{unmet}} = \frac{\sum_t U_t}{\sum_t Q_t}$, which is the primary comfort metric.

In addition, we use a standardised weekly diagnostic plot for qualitative comparison, showing tank temperature and failures, applied action, $\lambda(t)$ with draw markers, draw and unmet volumes, and per-step reward for the PPO policy.

Table 2: Weekly performance over 20 stochastic demand realisations, reported as mean (standard deviation). Note: Always-On is an extreme reference strategy (not a deployable baseline); see Section 3.4.1.

Controller	Energy (kWh)	Unmet volume (L)	Unmet fraction	Switches
Always-On	504.0 (0.0)	65 (46)	0.076 (0.053)	0.0 (0.0)
Thermostat	162.6 (6.2)	111 (56)	0.131 (0.063)	214.5 (4.7)
Timer	143.3 (5.3)	369 (118)	0.432 (0.125)	139.9 (4.4)
Risk-Band	153.8 (6.0)	152 (55)	0.179 (0.060)	176.6 (5.4)
PPO	254.0 (4.8)	53 (46)	0.062 (0.052)	329.7 (4.7)

4 Results and Discussion

This section compares the PPO controller against deployable baselines under identical stochastic demand realisations. The aim is to separate the value of probabilistic demand information from the value of reinforcement learning as an optimisation method. Unless stated otherwise, results are aggregated over $N = 20$ evaluation seeds, each representing one simulated week.

4.1 Controllers and evaluation summary

We evaluate Always-On, Thermostat, Timer, Risk-Band, and PPO as defined in Section 3.4. Risk-Band and PPO both have access to the same demand-risk signal $\lambda(t)$. Differences between them therefore reflect the effect of policy optimisation and representational flexibility rather than access to additional information.

4.2 Aggregate performance and trade-offs

Table 2 reports the primary engineering metrics: weekly energy consumption, volume-weighted comfort failures, and switching. Values are shown as mean (standard deviation) over 20 seeds, rounded to a consistent precision to support comparison.

The baselines illustrate the two extremes. Always-On maintains relatively low unmet fraction but at very high energy cost. Timer reduces energy slightly relative to Thermostat, but performs poorly on comfort because a fixed schedule is brittle under variable-duration events. Thermostat provides a moderate compromise but does not condition heating decisions on demand risk.

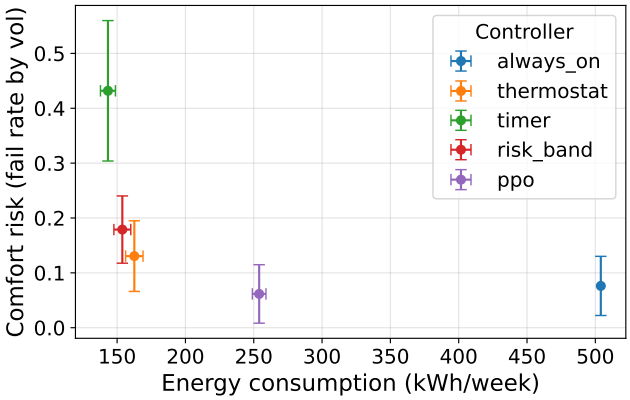

Figure 2: Energy–comfort trade-off. Points denote mean performance; error bars show one standard deviation across seeds.

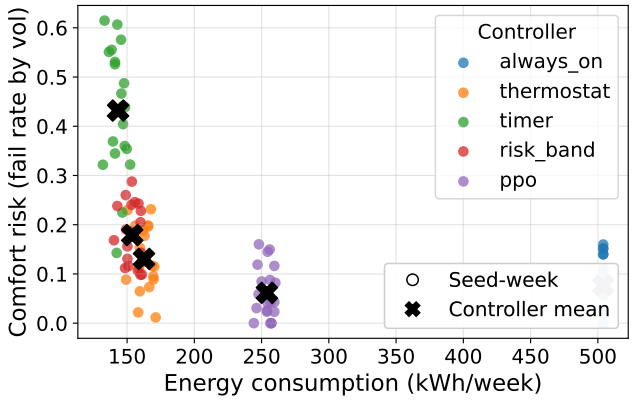

Figure 3: Per-seed energy consumption versus unmet fraction. Each point corresponds to one seed-week; crosses denote controller means.

Risk-Band demonstrates that exploiting $\lambda(t)$ yields a clear improvement over Timer and Thermostat at similar energy levels, supporting the value-of-information framing. PPO achieves the lowest average unmet fraction and unmet volume, but does so with substantially higher energy and markedly higher switching. This indicates that, in this setup, reinforcement learning provides incremental comfort refinement rather than a dominant shift of the energy–comfort frontier, and that the refinement comes with actuation and energy costs.

Figure 2 summarises the energy–comfort trade-off using mean and one standard deviation across seeds. Timer occupies a low-energy, high-failure region while Always-On lies in a high-energy, lower-failure region. Risk-Band shifts the trade-off by responding to demand risk. PPO attains lower failure on average than Risk-Band, but at a higher energy operating point.

4.3 Variability across stochastic demand realisations

Figure 3 shows per-seed outcomes, which makes week-to-week variability explicit. Risk-Band exhibits relatively concentrated performance across seeds, whereas PPO shows broader dispersion. In deployment terms, this matters because occasional poor weeks are user-visible as cold-water events. The spread suggests that PPO's behaviour is more sensitive to particular demand realisations, while the heuristic controller is more predictable.

4.4 Failure rates conditioned on demand risk

Figure 4 reports volume-weighted failure rate as a function of the risk signal $\lambda(t)$, binned into deciles. Failures concentrate in higher-risk bins across controllers, as expected. Risk-Band reduces failures relative to Thermostat during high-risk periods by allocating additional thermal headroom. PPO further reduces failure rates in the highest-risk bins, consistent with learning a stronger anticipatory response, but the qualitative pattern matches the risk-aware heuristic.

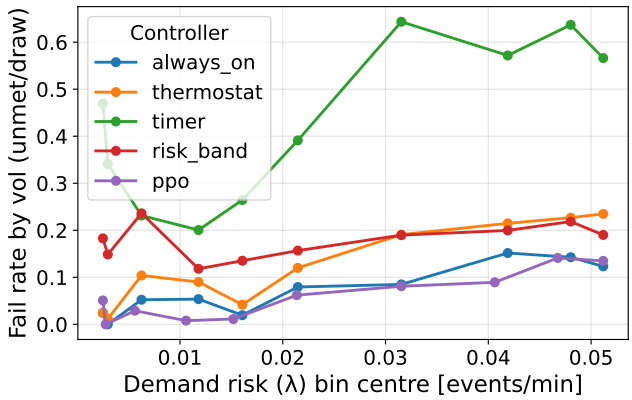

Figure 4: Volume-weighted failure rate as a function of demand risk $\lambda(t)$, binned by deciles.

4.5 Temperature allocation as a function of risk

Figure 5 compares tank temperature distributions conditioned on demand-risk bins for PPO and Risk-Band. Risk-Band shows discrete operating regimes consistent with switching between two setpoints. PPO exhibits a smoother progression across risk bins, indicating that it learns a graded mapping from risk to temperature. This behavioural difference is consistent with PPO's lower mean unmet fraction, but Table 2 shows that the improvement is achieved at notably higher energy and switching.

4.6 Actuation cost and switching behaviour

Figure 6 plots switching count against energy consumption. PPO incurs substantially higher switching than Risk-Band, even with dwell-time constraints. This is a practical concern for domestic immersion heaters because it directly relates to relay wear and noise, and it may be unacceptable even if comfort improves. Risk-Band provides a more balanced operating regime, with lower switching at a lower energy cost.

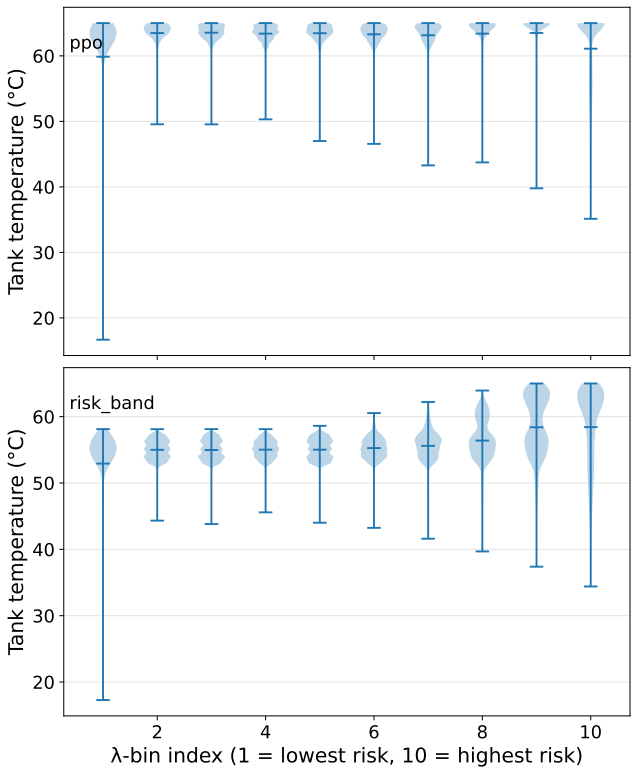

Figure 5: Tank temperature distributions conditioned on demand risk for PPO and Risk-Band.

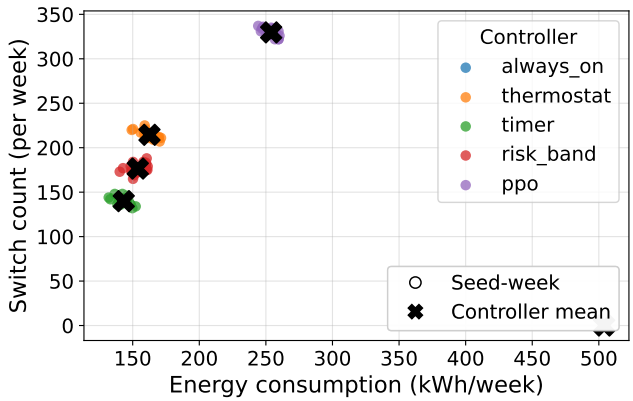

Figure 6: Weekly switch count versus energy consumption across controllers.

4.7 Representative-week behaviour

To ground the aggregate metrics in behaviour, Figure 7 shows one representative week for PPO and Risk-Band under the same demand realisation. Unmet draw events appear as markers on the temperature trajectories. The corresponding weekly risk profile for the demand model is illustrated in Figure 1, showing how peaks in $\lambda(t)$ align with clusters of draw-active steps.

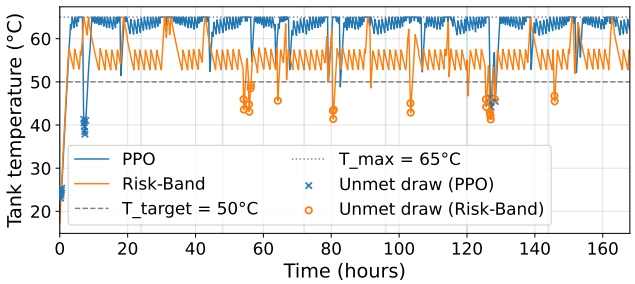

Figure 7: Representative weekly tank temperature trajectories for PPO and Risk-Band under the same demand realisation. Unmet draw events are indicated by markers.

Risk-Band maintains a lower, more regular temperature pattern and incurs unmet events during prolonged or clustered demand. PPO maintains a higher buffer more often, reducing unmet events but increasing energy use and switching. This aligns with the aggregate results: PPO improves comfort on average, but at the cost of operating closer to a conservative strategy.

4.8 Summary

Across all evaluations, three observations are consistent:

- Demand-risk information is the primary driver of improved energy–comfort performance. Risk-Band substantially improves over Timer and Thermostat at similar energy levels.
- PPO further reduces comfort failures on average, but does so at higher energy consumption and substantially higher switching.
- For deployable immersion-heater control, simple risk-aware heuristics capture most of the attainable benefit while providing more predictable actuation behaviour.

5 Conclusion

This paper examined the control of domestic hot-water systems under stochastic, duration-aware demand using a physics-based simulation environment. Hot-water usage was modelled as a non-homogeneous Poisson process with multi-step draws, allowing controllers to operate under realistic uncertainty and binary actuation constraints. By benchmarking a reinforcement learning controller against conventional baselines and a risk-aware heuristic that exploits the same demand-intensity signal, we explicitly separated the value of probabilistic demand information from the value of algorithmic complexity.

Across multiple stochastic demand realisations, results show that access to a real-time demand-risk signal $\lambda(t)$ accounts for the majority of achievable improvements in the energy–comfort trade-off. A simple risk-aware heuristic captures most of these gains while exhibiting lower variance and reduced switching activity. Reinforcement learning provides incremental refinement by learning smoother anticipatory behaviour, but at the cost of increased complexity and less predictable actuation.

These findings suggest that for near-term deployment in residential immersion-heater systems, transparent risk-aware heuristics

may offer a more practical balance of efficiency, robustness, and interpretability than black-box learning-based controllers. The study employs a simplified thermal model and a step-wise delivered-temperature proxy, and therefore does not capture stratification or outlet-layer dynamics that influence absolute comfort in real cylinders. Results should be interpreted primarily as a comparative analysis of controller behaviour and information value under consistent assumptions. Incorporating low-order stratified models and established optimisation-based baselines such as model predictive control represents an important direction for future work. Extensions to time-of-use pricing, renewable generation, and longer-horizon coordination may further clarify when learning-based approaches provide additional value.

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